

ACRME Placement Scoring Weights — Complete Reference

Project: Azure Capacity Reservation Management Engine (ACRME)

Classification: Technical Reference

Version: 1.1

Date: August 2026 (reconciled to Requirements Baseline v2.4, 7 Sep 2026)

v2.4 reconciliation note (7 Sep 2026). The five placement-scoring weights (α , β , γ , δ , ϵ) and their environment-specific formulas are **unchanged** by Baseline v2.4 — no weight value or formula is revised here. v2.4 clarifies one adjacency that this document must not conflate: the **ϵ zone-diversity signal** below (a region-selection weight, `az_count / 3`, PLC-007) is **distinct from PLC-011**, the new **even intra-region zone-distribution target** (`$\approx 1 / \text{zone\_count}$` per zone) with rebalancing. ϵ nudges region ranking toward regions that have more zones; **PLC-011 governs how a workload's VMs are spread across the zones of the region already chosen**, and can trigger a rebalancing action when zone skew exceeds the configured tolerance (C-13, formula Appendix A.9). See the ϵ section below for the full reconciliation. The remaining v2.4 additions (CAP-020..024, OPS-006) do not change any scoring formula.

Overview

The ACRME placement engine uses **three environment-type-specific scoring formulas** (PS_Prod, PS_NonProd, PS_DR) to rank Azure regions for Prod, NonProd/CVAL, and DR environment placement. All three formulas share the same **default weight values**:

```
 $\alpha$  = 0.30 (CRG Capacity)
 $\beta$  = 0.20 (Quota Headroom)
 $\gamma$  = 0.25 (Distribution Fairness)
 $\delta$  = 0.15 (DR/Overflow Health)
 $\epsilon$  = 0.10 (Zone Diversity)
```

However, **α** and **δ** have **different semantic meanings** per environment type — they measure different signals depending on whether you're scoring a Prod, NonProd, or DR region.

Weight Definitions by Component

α (Alpha) — CRG Capacity Signal [Weight: 0.30]

Purpose: Measures available capacity in the region's CRG most relevant to the environment being placed.

Semantic per environment:

Environment	α Measures	Formula
PS_Prod	NonProd CRG overflow headroom relative to Prod CRG size. High α = the associated NonProd CRG has capacity to absorb overflow if Prod grows.	<code>non-prod_crg_effective_free / prod_crg_quantity</code>
PS_NonProd	Fraction of NonProd CRG that is truly available (free_slots minus dr_overflow_reserve). This is the primary capacity signal for NonProd placement.	<code>non-prod_crg_effective_free / nonprod_crg_quantity</code>
PS_DR	Fraction of DR CRG capacity that is uncommitted. High α = more room for additional DR assignments.	<code>dr_crg_free_slots / dr_crg_quantity</code>

Why α is the highest weight (0.30): Capacity is the most critical constraint. A region with zero headroom cannot accept new placements regardless of quota or distribution. α is highest when capacity is tight; can be reduced to 0.20 once regions reach steady-state utilization (40–60%).

β (Beta) — Quota Headroom [Weight: 0.20]

Purpose: Measures available Azure subscription/group quota headroom for the relevant CRG.

Semantic per environment:

Environment	β Measures	Formula
PS_Prod	Prod group quota headroom ratio	<code>prod_crg_quota_headroom_vcpu / prod_crg_quota_limit_vcpu</code>
PS_NonProd	NonProd floor-adjusted quota headroom (effective ceiling minus used)	<code>non-prod_crg_quota_headroom_vcpu / non-prod_crg_quota_limit_vcpu</code>
PS_DR	DR quota headroom within the DR floor allocation	<code>dr_crg_quota_headroom_vcpu / dr_crg_quota_limit_vcpu</code>

Why $\beta = 0.20$: Quota exhaustion silently blocks CR creation and VM deployment. β ensures the engine directs placements toward regions with quota headroom. Lower than α because quota can be requested (takes time but is not a hard ceiling like physical capacity).

Note: All β calculations read from **quota-group-level fields** (two-group model: Prod-only group, shared NonProd+DR group), not legacy per-subscription quota. See ADR-002.

γ (Gamma) — Distribution Fairness [Weight: 0.25]

Purpose: Acts as a **self-correcting load balancer**. Rewards regions with fewer customers already placed, driving even distribution across regions over time.

Semantic (same for all environments):

$$\gamma = 1 - (\text{customer_count_in_this_region} / \text{total_customers})$$

Environment	γ Measures	Formula
PS_Prod	Prod region fairness	$1 - (\text{consumer_count_prod} / \text{total_customers})$
PS_NonProd	NonProd region fairness	$1 - (\text{consumer_count_nonprod} / \text{total_customers})$
PS_DR	DR region fairness	$1 - (\text{consumer_count_dr} / \text{total_customers})$

Why $\gamma = 0.25$: Equal to capacity in importance. As a region accumulates customers its γ score drops, redirecting future placements away from it until all regions reach roughly equal count (steady-state expected count per region per env type = $\text{total_customers} / 3$). Increase γ toward 0.35 if score convergence results in uneven loading; decrease toward 0.15 if capacity constraints should dominate fairness.

Note: The PRR (Section 28) updates Distribution to use **demand units** rather than customer count:

$$1 - (\text{Region_Assigned_Demand} / \text{Total_Assigned_Demand})$$

δ (Delta) — DR/Overflow Health [Weight: 0.15]

Purpose: Environment-specific **resilience signal** — measures DR coverage readiness (Prod/DR) or overflow capacity health (NonProd).

Semantic per environment:

Environment	δ Measures	Formula	Intent
PS_Prod	DR CRG coverage readiness of this region's own DR CRG. When scoring a Prod region, δ captures the health of the region's DR CRG — a Prod region whose DR CRG is well-covered scores higher.	<code>dr_crg_coverage_ratio</code>	Prod placement favors regions with healthy DR capacity
PS_NonProd	Overflow capacity health — intentionally shares the same ratio as α . Rewards regions genuinely under-utilised from both a NonProd-placement and a DR-overflow perspective.	<code>non-prod_crg_effective_free / non-prod_crg_quantity</code> (same as α)	NonProd placement prioritizes regions with DR overflow headroom
PS_DR	Coverage ratio health relative to target. Scores 1.0 when coverage $\geq$ <code>dr_ratio_max</code> (0.40). Proportionally below 1.0 for coverage between <code>dr_ratio_min</code> (0.30) and <code>dr_ratio_max</code> (e.g. $0.35/0.40 = 0.875$).	<code>min(1.0, dr_crg_coverage_ratio / dr_ratio_max)</code>	DR placement drives buffer margins toward target

Why δ is different per env: Each environment has a distinct resilience need. Prod cares about its paired DR region's readiness; NonProd cares about overflow headroom it can yield to DR during a crisis; DR cares about its own coverage health.

Why $\delta = 0.15$: Lower than $\alpha/\beta/\gamma$ because it's a forward-looking health signal rather than an immediate placement blocker. Increase δ toward 0.20 if DR buffer violations appear; decrease if capacity/distribution should dominate.

Trade-off (PS_NonProd): δ duplicates α intentionally — this was flagged in the PRR (Section 28) as double-counting. The corrected pilot formula proposes revised NonProd weights: `PS_NonProd = 0.35·Capacity + 0.25·Quota + 0.25·Distribution + 0.05·DR_Overflow_Integrity + 0.10·Zones` to reduce the duplication.

ε (Epsilon) — Zone Diversity [Weight: 0.10]

Purpose: Rewards regions with more availability zones, ensuring placed workloads can leverage Azure's zone-redundancy features.

Semantic (same for all environments):

$$\epsilon = \text{az_count} / 3$$

Normalizes to 1.0 for a 3-zone region (maximum diversity); proportionally lower for 2-zone (0.667) or 1-zone (0.333) regions.

Environment	ε Measures	Formula
All (PS_Prod, PS_NonProd, PS_DR)	Zone diversity	<code>az_count / 3</code>

Why ε = 0.10 (lowest weight): Zone diversity is desirable but not a placement blocker. Hard constraint **HC-5 ZONE_AVAILABILITY** already enforces a minimum zone count — ε acts as a tiebreaker that favors 3-zone regions when all else is equal. Increase ε toward 0.15 if zone diversity is a critical operational requirement; keep at 0.10 if capacity/distribution dominate.

ε (region-selection) vs. PLC-011 (intra-region even distribution) — v2.4 reconciliation

ε and PLC-011 both concern availability zones but operate at **different stages** and must not be conflated:

Aspect	ϵ — Zone Diversity (this weight, PLC-007)	PLC-011 — Even Zone-Distribution Target
Stage	Region selection (which region wins)	Zone placement within the already-selected region
Signal / target	$\epsilon = \text{az_count} / 3$ — a soft ranking signal	Even spread $\approx 1 / \text{zone_count}$ per zone ($\approx 33\%$ each in a 3-zone region) — a placement target
Effect	Nudges score toward regions that have more zones	(a) Prefers the under-represented zone for new placement; (b) raises a re-balancing recommendation/action when skew exceeds tolerance
Trigger	None — tiebreaker only	Max Skew > Configured Skew Tolerance (C-13) triggers re-balance (subject to approval + Azure feasibility)
Formula	$\text{az_count} / 3$	Appendix A.9 (below)
CRG interaction	—	Keeps per-AZ CRG sizing under CAP-023 balanced

The baseline is explicit that PLC-011 “is a placement **target**, not merely the zone-diversity scoring signal used in PLC-007 weighting ... without an explicit target the ϵ zone-diversity term only nudges scoring and cannot trigger a rebalance.” ϵ therefore remains purely a **region-ranking** tiebreaker; the intra-region distribution target and rebalancing are governed separately by PLC-011.

Appendix A.9 — Even zone-distribution skew formula (PLC-011):

Even Zone Share 3-zone region)	= $1 / \text{Zone Count}$	(e.g. $1/3 \approx 33\%$ in a
Target VMs per Zone	= $\text{round}(\text{Workload VM Count} / \text{Zone Count})$	
Zone Skew(z)	= $\text{VMs in zone } z - \text{Target VMs per Zone}$	
Max Skew	= MAX over zones z of $ \text{Zone Skew}(z) $	
Rebalance Trigger	= $\text{Max Skew} > \text{Configured Skew Tolerance (C-13)}$	
Preferred Placement Zone	= $\text{argmin over zones } z \text{ of } (\text{VMs in zone } z)$	# under-represented zone

The target ratio and tolerance are configuration-driven (`PlacementPolicy` , constraint **C-13**). See the baseline PLC-011 and Appendix A.9, and ADR-003 (zone-distribution interaction) for the DR-sizing cross-reference.

Complete Formula Definitions

PS_Prod(r) — Production Region Score

```

PS_Prod(r) =
  0.30 × (nonprod_crg_effective_free / prod_crg_quantity)           ← α: NonProd
overflow headroom
  + 0.20 × (prod_crg_quota_headroom_vcpu / prod_crg_quota_limit_vcpu) ← β: Prod quota
headroom
  + 0.25 × (1 - consumer_count_prod / total_customers)             ← γ: Prod dis-
tribution fairness
  + 0.15 × dr_crg_coverage_ratio                                     ← δ: DR readi-
ness of likely DR target
  + 0.10 × (az_count / 3)                                           ← ε: zone di-
versity

```

Guards:

- If `prod_crg_quantity == 0` → `PS_Prod = 0.0`
- If `prod_crg_quota_limit_vcpu == 0` → `β term = 0.0`
- If `dr_crg_coverage_ratio is None` → `δ term = 0.0` (no DR CRG established yet)

PS_NonProd(r) — Non-Production / CVAL Region Score

Current formula (with intentional α/δ duplication):

```

PS_NonProd(r) =
  0.30 × (nonprod_crg_effective_free / nonprod_crg_quantity)       ← α: ef-
fective NonProd headroom
  + 0.20 × (nonprod_crg_quota_headroom_vcpu / nonprod_crg_quota_limit_vcpu) ← β: Non-
Prod quota headroom
  + 0.25 × (1 - consumer_count_nonprod / total_customers)         ← γ: Non-
Prod distribution
  + 0.15 × (nonprod_crg_effective_free / nonprod_crg_quantity)     ← δ: over-
flow capacity health (same as α)
  + 0.10 × (az_count / 3)                                           ← ε: zone
diversity

```

Proposed pilot formula (corrected to avoid double-counting):

```

PS_NonProd(r) =
  0.35 × (nonprod_crg_effective_free / nonprod_crg_quantity)       ← α: capa-
city (increased)
  + 0.25 × (nonprod_crg_quota_headroom_vcpu / nonprod_crg_quota_limit_vcpu) ← β: quota
headroom
  + 0.25 × (1 - consumer_count_nonprod / total_customers)         ← γ: dis-
tribution
  + 0.05 × (nonprod_crg_effective_free / nonprod_crg_quantity)     ← δ: DR
overflow integrity (reduced)
  + 0.10 × (az_count / 3)                                           ← ε: zone
diversity

```

Guards:

- If `nonprod_crg_quantity == 0` → `PS_NonProd = 0.0`
- If `nonprod_crg_quota_limit_vcpu == 0` → `β term = 0.0`

PS_DR(r) — Disaster Recovery Region Score

PS_DR(r) =

0.30 × (dr_crg_free_slots / dr_crg_quantity)

CRG headroom

+ 0.20 × (dr_crg_quota_headroom_vcpu / dr_crg_quota_limit_vcpu)

quota headroom

+ 0.25 × (1 - consumer_count_dr / total_customers)

distribution

+ 0.15 × min(1.0, dr_crg_coverage_ratio / dr_ratio_max)

age ratio health

+ 0.10 × (az_count / 3)

diversity

← α: DR

← β: DR

← γ: DR

← δ: cover-

age ratio health

← ε: zone

Guards:

- If dr_crg_quantity == 0 → PS_DR = 0.0 (no DR CRG — region ineligible)
- If dr_crg_quota_limit_vcpu == 0 → β term = 0.0
- If dr_crg_coverage_ratio is None → δ term = 0.0

Weight Summary Table

Component	Symbol	Weight	PS_Prod Measures	PS_NonProd Measures	PS_DR Measures
CRG Capacity	α	0.30	NonProd overflow headroom	NonProd effective headroom	DR CRG free capacity
Quota Headroom	β	0.20	Prod group quota ratio	NonProd floor-adjusted quota	DR quota within floor
Distribution	γ	0.25	Prod region fairness	NonProd region fairness	DR region fairness
DR/Overflow Health	δ	0.15	DR coverage readiness	Overflow capacity (same as α)	Coverage vs target ratio
Zone Diversity	ε	0.10	az_count / 3	az_count / 3	az_count / 3

Clamping & Score Range

Every raw component ratio is **clamped** before weighting:


```
Clamp(x) = max(0, min(1, x))
```

This ensures each component contributes a value in the range `[0.0, 1.0]`, and the final weighted score for any region is:

```
PS(r) ∈ [0.0, 1.0]
```

A uniform jitter of ± 0.01 is added as a tiebreaker.

Tuning Guidance

When to adjust α (Capacity)

- **Increase α toward 0.35–0.40** when capacity is tight and regions are unevenly provisioned.
- **Reduce α toward 0.20** once regions reach steady-state utilization (40–60%) and distribution becomes the primary concern.

When to adjust γ (Distribution)

- **Increase γ toward 0.35** if score convergence results in uneven region loading (some regions accumulate too many customers).
- **Decrease γ toward 0.15** if capacity constraints should dominate fairness (tight capacity environments).

When to adjust δ (DR/Overflow Health)

- **Increase δ toward 0.20** if DR buffer violations or coverage gaps appear frequently.
- **Reduce δ toward 0.10** if capacity/distribution should dominate and DR health is consistently good.

When to adjust ϵ (Zone Diversity)

- **Increase ϵ toward 0.15** if zone diversity is a critical operational requirement (e.g., all placements must strongly favor 3-zone regions).
- **Keep ϵ at 0.10** (default) if capacity, quota, and distribution are the dominant concerns.

Default recommendation: Start with the default weights ($\alpha=0.30$, $\beta=0.20$, $\gamma=0.25$, $\delta=0.15$, $\epsilon=0.10$) for the pilot. After 30–50 customer placements, analyze the score breakdown data in `OperationRecord` and adjust incrementally based on observed convergence behavior.

Source References

- **Multi-Region Placement Design** (`docs/research/multi_region_placement_design.md`) — lines 449–531, full formula definitions + weight table
- **Design Change Summary** (`docs/research/design_change_summary.md`) — CR/CRG-7, lines 153–193, formula rationale
- **Production Readiness Review** (`docs/ac-rme_production_readiness_review_and_architecture.md`) — Section 28 lines 1095–1160, corrected scoring model

- **ADR-001 Region Selection** (docs/adr/acrme_adr_001_region_selection.md) — corrected scoring model section
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Evidence Tags

- [Documented] — Defined in Multi-Region Placement Design and PRR Section 28
 - [Decided] — Decision Log D9 (env-type-specific formulas)
 - [Assumed] — Default weights retained for pilot comparison; revised weights proposed but not empirically validated
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Document Status: Reference

Next Review: After pilot Phase 4 (policy tuning with 30–50 placements measured)